



Competitive protein adsorption in isocratic anion exchange chromatography

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ABSTRACT

Protein chromatography is the dominant method of purification of biopharmaceuticals. Although all practical chromatography involves competitive absorption and separation of M. species, competitive protein absorption has remained inadequately understood. We previously introduced the measurement of equilibrium protein adsorption isotherms with all intensive variables held constant, including competitor concentration. In this work, we introduce isocratic chromatographic retention measurements of dynamic protein adsorption in the presence of a constant concentration of a competitor protein. These measurements are achieved by establishing a dynamic equilibrium with a constant concentration of competitor (insulin) in the mobile phase flowing through an ion exchange adsorbent column and following the behavior of a test protein (α -lactalbumin) injected into this environment. We observed decreased retention times for α -lactalbumin in presence of the competitor. The presence of competitor also reduces the heterogeneity of the sites available for adsorption of the test protein. This investigation provides an approach to fundamental understanding of competitive dynamics of multicomponent protein chromatography.

1. Introduction

Ion-exchange chromatography is ubiquitous in the purification of biopharmaceutical proteins [1–5]. The factors that govern protein adsorption are the intrinsic affinity of the protein for the adsorbent under the operating conditions, and competition from other proteins [6–8]. Protein adsorption is usually described in terms of single-component isotherms generated using batch adsorption equilibrium and frontal chromatography, and is modeled using a variety of empirical and mechanistic approaches [9–14]. Since all practical applications of chromatography involve separation of multicomponent systems, single-component isotherms have been extended to multicomponent system to address competitive multicomponent adsorption, although with limited success [15–22].

In essentially all previous work, study of competitive adsorption obtained from multicomponent (binary) systems was complicated by simultaneous variation of the concentrations of each of the two proteins.

Cano et al. (2005) provided the first competitive adsorption isotherms with all intensive variables held constant, including competitor concentration. To our knowledge, however, dynamic competitive binding behavior in protein chromatography has never been studied under conditions of constant competitor concentration. Such investigations are complicated by the fact that batch injection of a mixture onto a chromatographic column normally results in separation of the test protein from the competitor as they move along the column, confounding interpretation of the effects of the competitor on the chromatographic behavior of the test protein. In this work, we introduce the first isocratic retention approach to measure dynamic protein chromatographic behavior in the presence of a constant concentration of a competitor protein. This was achieved by establishing a dynamic equilibrium with a constant concentration of competitor in the mobile phase of an ion exchange chromatographic column and following the behavior of a test protein injected into this environment.

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2. Materials and methods

2.1. Proteins

We used insulin (5.8 kDa; R_h , 1.19 nm; Cat # I0908, Sigma-Aldrich) and apo α -lactalbumin (14.3 kDa; R_h , 1.43 nm; Cat # L6010, Sigma-Aldrich) as competitor and test proteins, respectively. This choice was based on the similar physicochemical properties of these proteins and our earlier work involving the anionic charge cluster created on “apo” α -lactalbumin (pI 5.0) by Ca^{+2} depletion [23,24]. Insulin (pI 5.3) has a hydrodynamic radius comparable to our test protein, α -lactalbumin. Also, insulin is not prohibitively expensive in the large quantities required and did not produce the column clogging observed in initial experiments with bovine serum albumin.

2.2. Alpha-lactalbumin labeling

Alexa 555 labeling of Ca^{+2} -depleted α -lactalbumin (α -La) was performed as previously described [24] and enabled its sensitive and selective quantitation in the presence of excess insulin by 555 nm absorbance. Unreacted dye was removed using a Zeba™ Spin Desalting Column (5 mL, Cat. #89,891, Thermo Fischer Scientific), followed by size-exclusion chromatography (SEC) on a Superdex 75 10/300 GL (Cytiva) column on an ÄKTA Purifier 10 with phosphate-buffered saline (10 mM sodium phosphate buffer, 150 mM sodium chloride, pH 7.4) as the mobile phase. Pooled fractions containing labeled protein were further purified using a DEAE Sepharose FF (Cytiva) column (41 mm height $\times$ 5 mm inner diameter) with a 0.85-mL column volume (CV) pre-equilibrated with 10 CV of 25 mM Tris, pH 8.0 (“buffer A”). After 500 μ L loading, the column was washed with 5 CV of buffer A, then eluted with a 40 CV linear gradient of 0 - 1.0 M sodium chloride in buffer A. Fractions with dye-to-protein ratio of 1.0 ± 0.1 as assessed by absorbance at 280 nm and 555 nm were pooled and analyzed using MALDI-TOF as previously described, to confirm single labeling of α -lactalbumin [23].

2.3. Adsorbent preparation

Spermine Sepharose was prepared as discussed by Dhamane et al. [25], with slight modifications. Briefly, Sepharose CL-6B (500 μ L settled resin in 1.5 mL water) was treated with 200 mM sodium periodate for 2 h at room temperature. Aldehyde-activated Sepharose was washed with deionized water, then resuspended in 50 mM sodium borate buffer, pH 9.2. To 500 μ L of activated Sepharose CL-6B resin, 160 μ L of 60 mM spermine tetrahydrochloride solution was added and the final volume adjusted to 2.0 mL with borate buffer. This suspension was allowed to react for 75 min at ambient temperature on a rotator followed by centrifugation for 2 min at $16,000 \times g$. The settled spermine Sepharose was treated with 2.0 mL of 2 mM sodium borohydride in 50 mM borate buffer to reduce the initially formed Schiff base to a stable secondary amine linkage. Spermine Sepharose was washed with 20 mM phosphate buffer, pH 7.0, and stored at 4 °C. The ligand density of spermine Sepharose was determined by acid-base titration as described by Dhamane et al. (2014). Briefly, the adsorbent was acidified with 0.1 N hydrochloric acid and then 2.5 mM sodium hydroxide passed through the column to reach the base equivalents required to fully titrate the spermine, as detected by in-line pH probe on the ÄKTA Purifier.

2.4. Batch adsorption isotherm measurements

Competitive batch equilibrium protein adsorption isotherms were measured by varying the concentration of the target protein α -La while maintaining constant competitor (insulin) concentration. Aliquots (15 μ L) of 70 % (v/v) spermine Sepharose suspension in 10 mM Tris, 25 mM NaCl, pH 8.0 (“binding buffer”) with or without insulin competitor were transferred to 1.5 mL low protein-binding tubes (Cat. #90,411, Thermo

Scientific). The spermine Sepharose was pre-equilibrated with 1 mL of the test buffer (binding buffer containing the desired concentration of insulin) for 120 min. After incubation, the tubes were centrifuged for 10 min at $10,000 \times g$ and the supernatant discarded. The adsorbent was again resuspended in the same test buffer as before, followed by the addition of α -La (to 1, 2.5, 5, 10, 20, 30 and 50 μ g/mL) from a stock solution prepared in binding buffer, and the volume adjusted to 1 mL using test buffer. The tubes were placed on a gyratory rotator at room temperature for 1 h (a time determined in preliminary experiments to be sufficient for equilibration), then centrifuged. The supernatant was carefully decanted, and absorbance measured at 555 nm. The adsorbent pellets were washed by resuspension in 1 mL of binding buffer followed by centrifugation, then resuspended for elution for 1 h in 1 mL of 10 mM Tris, 1 M NaCl, pH 8.0 (“elution buffer”). The α -La concentrations in the supernatant and eluate were quantified by 555 nm absorbance using a Tecan Infinite M200 Pro (Tecan, USA).

2.5. Competitive isocratic chromatography

Column chromatography experiments were performed using a glass column (41 mm height $\times$ 5 mm inner diameter) packed with spermine Sepharose to a CV of 0.85 mL. A dynamic equilibrium was established by equilibrating the column with least 40 CV of 5 mM or 10 mM Tris-HCl buffer pH 8.0, with or without added insulin, to a final concentration of 0, 0.6, 6, 60 or 300 μ g/mL at a linear flow velocity of 180 cm/h. A single 10 μ L pulse of 3.8 μ g/mL of α -La was injected onto the column using an injection loop followed by isocratic elution at 180 cm/h.

3. Data analysis

The key innovation of this work is that the concentration of the competitor protein was for the first time treated as a constant, intensive variable in chromatography. Single-component models were utilized to generate isotherms to compare to competitive binary equilibrium data. This simplification allowed us to evaluate the influence of various insulin concentrations on the α -La-spermine Sepharose interaction. The equilibrium adsorption data were analyzed using the Langmuir–Freundlich (L–F) isotherm as discussed previously by Cano et al. [18].

The Langmuir–Freundlich model is

$$q^* = \frac{q_m (C^*)^n}{K_d + (C^*)^n} \quad (1)$$

where n is the heterogeneity index, with low values of n indicating high surface heterogeneity, and n approaching unity indicating a homogeneous binding surface, equivalent to the classic Langmuir model; K_d is the apparent dissociation constant; C^* is the equilibrium concentration of the protein in the solution; q^* is the amount of protein bound to the adsorbent at equilibrium; and q_m is the maximum capacity of the adsorbent for the protein. The Langmuir–Freundlich isotherm parameters (q_m , K_d , and n) were obtained by plotting the experimental values for q^* (y-axis) as a function of C^* (x-axis) followed by fitting to the isotherm equation using non-linear regression analysis in GraphPad Prism software.

4. Results and discussion

Protein anion-exchange behavior is often described in terms of net charge, but this neglects the inherent charge heterogeneity of protein and adsorbent surfaces [26,27]. Heterogeneous protein surface charge distribution gives rise to charge clusters that preferentially interact with the adsorbent [26,27]. Similarly, there also are stochastic charge clusters on adsorbent surfaces. We have demonstrated the functional benefits of intentional pairing of charge clusters on the protein surface with an engineered clustered-charge ligand [24,28]. The choice of spermine ligand in this work was prompted by our previous studies of its

clustered-charge behavior and its higher affinity and capacity over conventional adsorbents of the same charge density [25].

4.1. Alpha-lactalbumin/insulin binary system

The multicomponent adsorption presented in this study was evaluated using low-ligand-density spermine Sepharose under constant competitor concentration. For the 1.75 mM spermine Sepharose used here, the ligand density on the porous stationary phase is low enough to yield an average inter-ligand distance of about 100 Å. The estimated Debye screening length of 20 Å at 25 mM salt would suggest that at this average spacing, inter-adsorbate electrostatic interactions would play only a small role in binding. Typical high-ligand-density adsorbents have crowded adsorption sites, and the small separation distance between two ligands eventually results in lateral adsorbate-adsorbate interactions. The use of low ligand density in this initial work avoids unintentional clustering of the spermine surface charges that might complicate interpretation of competitive adsorption data. Fig. 1 shows competitive adsorption isotherms generated with various concentrations of insulin competitor held constant over the entire range of the isotherm while offering increasing amounts of α -La. The maximum binding capacity (q_m) for α -lactalbumin was 732 μ g/mL of spermine Sepharose (1.75 mM) adsorbent. The adsorption of α -La on spermine-Sepharose was unaffected by 0.6 μ g/mL equilibrium concentration of insulin, but perturbed by all higher concentrations. Even at an equilibrium soluble α -La concentration of 45 μ g/mL, 6 μ g/mL insulin observably affects the adsorption of α -La present at seven times the insulin concentration. Interestingly, higher α -La concentrations only moderately reduce the competition from insulin (Fig. 1B; SI Fig. 1), though the effect of competitor is strongest at lower α -La concentrations.

Table 1 shows fitted Langmuir–Freundlich parameters K_d (apparent dissociation constant) and n (heterogeneity index) for the isotherms in Fig. 1. Insulin reduces the adsorbent affinity of α -La, evidenced by an overall increase in the value of K_d . At only 6 μ g/mL competitor, a noticeable reduction in affinity (higher K_d) was observed. In contrast, higher competitor loadings did not further reduce affinity. This, as well as the substantial effect of 6 μ g/mL competitor even at a great excess of α -La, suggests the presence of certain “favored sites” on the adsorbent for which insulin outcompetes α -La regardless of the concentration of the latter [23,24]. The heterogeneity index (n) increased moderately with incremental change in insulin concentration, implying that as the competitor concentration increases, the α -La experiences a more homogenous binding surface. One possible explanation for this behavior is that at 6 μ g/mL (or higher) concentration, insulin predominantly

Table 1

Langmuir–Freundlich parameters for α -lactalbumin/insulin system.

Insulin (μ g/mL, $n = 2$)	Langmuir–Freundlich parameters	
	K_d (μ g/mL)	n
0	12.02 ± 3.8	0.68 ± 0.13
0.6	11.05 ± 1.3	0.87 ± 0.10
6	20.14 ± 5.5	1.32 ± 0.19
60	24.27 ± 5.6	1.33 ± 0.15
300	21.4 ± 5.1	1.16 ± 0.17

occupies certain favored adsorption sites, thereby effectively reducing the number and variety of ligands available to α -La. The insulin-mediated blocking of certain adsorption sites reduces the binding surface heterogeneity, essentially removing those sites from the functional population of adsorption sites [29]. This observation is consistent with a narrowing of available site population under more stringent conditions of higher salt [30]. It also is consistent with our single-molecule spectroscopy study on the adsorption of α -La on spermine in the presence of competitor insulin [23].

4.2. Competitive isocratic chromatography

Isocratic chromatographic elution profiles were obtained by pulse injection of α -La onto a spermine Sepharose column operated isocratically with varying mobile-phase insulin competitor concentrations. α -La retention times decreased as the concentration of competitor in the mobile phase increased.

Fig. 2 shows the elution profiles obtained with spermine Sepharose adsorbents. On 0.5 mM and 1.75 mM spermine, retention time for α -La was prominently reduced by 0.018 μ g/mL and 0.6 μ g/mL of competitor insulin respectively. The retention time for α -La continued to decline at increasing concentrations of competitor. Moreover, for 1.75 mM spermine adsorbent, higher relative concentrations of competitor were required for effect compared to 0.5 mM spermine adsorbent. The significant effects of even low levels of competitor protein, also observed in equilibrium isotherm measurements, raise the possibility that these proteins compete for a relatively small number of favored adsorption sites; both our previous work [18] and our previous single-molecule observations [24,29,31] support this idea.

Table 2 presents a quantitative summary of the results shown in Fig. 2. Both adsorbent types show reduced peak tailing and asymmetry with increasing insulin competitor concentration. For 0.5 mM adsorbent, the peak width at half-maximum for α -La was noticeably reduced with increasing insulin concentration. A similar competition-induced

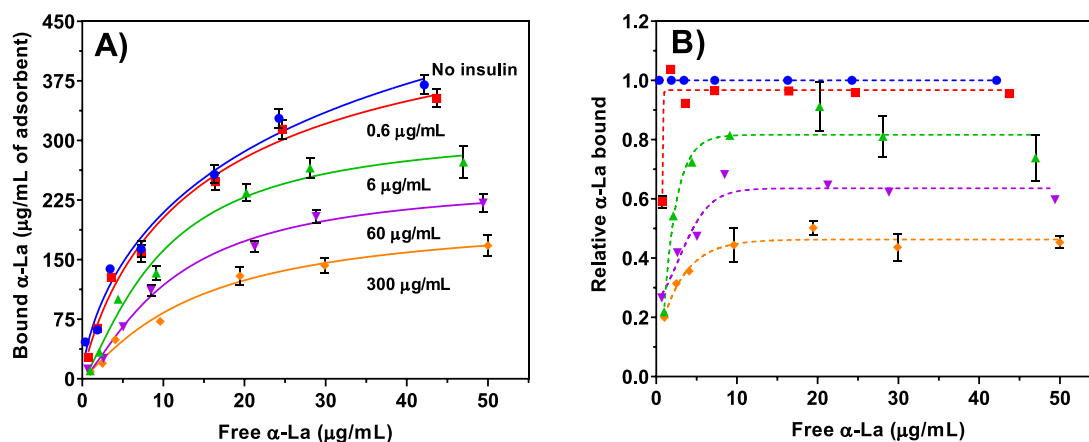


Fig. 1. (A) Competitive α -lactalbumin adsorption isotherms on 1.75 mM spermine Sepharose for varied constant concentrations of competitor insulin ($n = 2$). Solid lines are fits from the single-component Langmuir–Freundlich model with R^2 values ranging 0.979 to 0.9892. (B) α -lactalbumin bound in presence of competitor divided by that bound in the absence of competitor (“relative α -La bound”) vs. equilibrium α -La concentration (x-axis, $n = 2$). The dashed lines are to guide the eye; symbols and colors match those in (A).

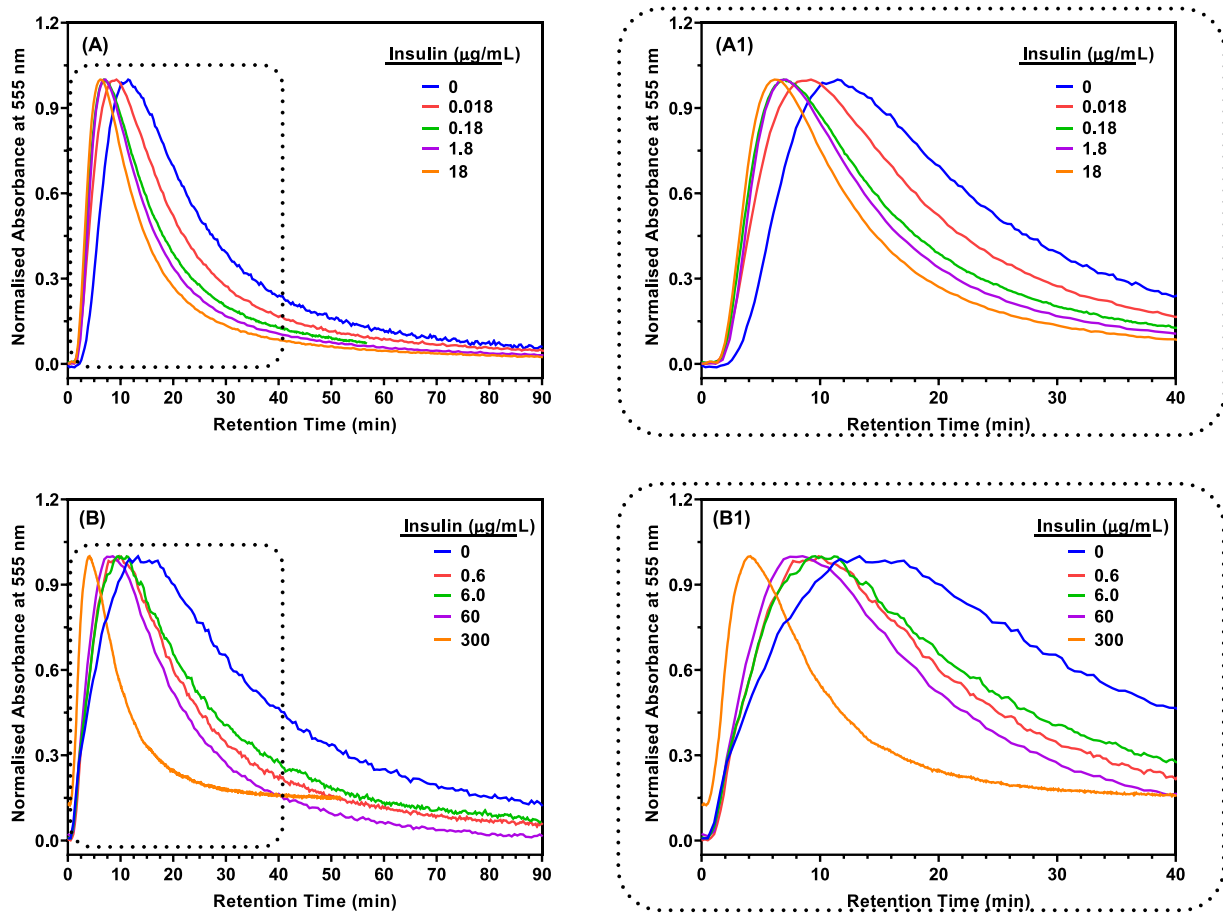


Fig. 2. Isocratic elution profiles for labeled α -lactalbumin from (A) 0.5 mM and (B) 1.75 mM spermine Sepharose column at several constant insulin concentrations. The graphs A1 and B1 are 0–40 min snapshots of elution profiles in A and B respectively. For 0.5 mM and 1.75 mM ligand density adsorbent 5 mM Tris–HCl, pH 8.0 and 10 mM Tris–HCl buffer pH 8.0 were used, respectively. The low ionic strength buffers were chosen to limit the salt screening induced at higher ionic strength. The absorbance signal for each elution profile was normalized using maximum peak absorbance for that profile. Elution profile of an unretained solute, acetone (1.5 min, not shown) was monitored at 280 nm.

Table 2
Peak maximum and average retention time and widths of peaks in Fig. 2.

Insulin ($\mu\text{g/mL}$)	Retention time at peak maximum (min)	Peak width at half-max ($w_{1/2}$)	Insulin ($\mu\text{g/mL}$)	Retention time (min) at peak maximum	Peak width at half-max ($w_{1/2}$)
0.5 mM spermine Sepharose			1.75 mM spermine Sepharose		
0	11.5	19.0	0	13.3	32.9
0.018	9.2	16.1	0.6	9.7	19.5
0.18	6.9	13.0	6	9.6	22.2
1.8	7.2	11.6	60	8.5	17.3
18	6.1	10.0	300	4.1	9.0

peak narrowing was also observed for 1.75 mM adsorbent at all offered insulin concentrations. The major source of peak tailing is heterogeneity in adsorption/desorption kinetics [32–34]. Therefore, the tails observed in the isocratic α -La elution profiles most likely arises from sites with longer desorption times. Our single-molecule experiments also observed the contributions of sites with longer desorption times, and similar observations of tailing due to adsorption site heterogeneity have been reported earlier [35,36]. As observed with the equilibrium adsorption isotherms in Fig. 1, higher insulin competitor concentrations may narrow the population of sites available to α -La, reducing both heterogeneity and peak tailing. We have previously reported evidence of competitors blocking adsorbate access to surface ligands in single-molecule experiments, and similar observations have been

reported elsewhere [23,37].
By employing equilibrium isotherm and competitive isocratic elution, we were able to demonstrate how clustered charge α -La-spermine Sepharose adsorption responded to the presence of a constant-concentration competitor. The results suggest that the effects of competitor blocking of otherwise preferred sites can be quite pronounced even in a dynamic system such as column chromatography. Use of higher ligand density adsorbents would yield ligand-crowded adsorbent that could result in lateral protein-protein interactions, possibly producing additional or averaged behaviors. Here, we concluded that site blocking likely played a dominant role in reducing retention and apparent heterogeneity at high competitor concentrations.

5. Conclusion

We investigated competitive adsorption by isocratic chromatograms with all intensive variables held constant, including, for the first time, the concentration of competitor. The deliberate use of low ligand density avoided inter-site interactions that could complicate interpretation of this initial competition data. We observed that the competitor reduces functional surface heterogeneity as well as apparent α -lactalbumin affinity for spermine adsorbent. In isocratic elution experiments, the mean retention time and peak width of α -lactalbumin are reduced with increased competitor concentration in the mobile phase. This study provides an approach to investigate the competitive dynamics of multicomponent protein chromatography.

CRedit authorship contribution statement

Ujwal Patil: Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Visualization, Writing – original draft, Writing – review & editing. **Katerina Kourentzi:** Methodology, Resources. **Richard C. Willson:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.chroma.2024.465114](https://doi.org/10.1016/j.chroma.2024.465114).

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